
$$\begin{aligned}
q''_{min} = & 0.0348 \left[\frac{c_{p, g} k_g^2}{\mu_g} \rho_g g(\rho_f - \rho_g) \right]^{0.549} [0.414 (\Delta T_{min, Cu})^{0.399} - 0.112]^{3.1217} \\
& \times \left[1 + 1.3 \left(\frac{\rho_f}{\rho_g} \right)^{0.2} \left(\frac{c_{pf} \Delta T_{sub}}{h_{fg}} \right) \right] [1 + 0.2 \sin(1.1 \theta)] \\
& \times \left[1 + 2.8 e^{-0.36 \left(\frac{L_c}{\lambda_d} \right)^2} \right] \left[\frac{k_{Cu}}{1 + k_w} \right]^{0.07}
\end{aligned}
\tag{21}$$
